

Akshat Patil

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EDUCATION

- Medhavi Skills University** Sep 2025 – Jun 2029
B.Tech in Computer Science & Engineering (AI/ML) Bengaluru, KA

EXPERIENCE

- Processing Foundation** Dec 2025 – Present
WebGL & WebGPU
 - p5.js Steward on the core team and WebGPU graphics steward, reviewing contributions and guiding project direction alongside maintainers
 - Engineered an instancing API (`instances(n).sphere()`) for the p5.strands WebGPU renderer, merged into core
 - Redesigned GPU compute dispatch to auto-spread large 1D workloads across 2D dimensions, enabling 1M+ particle simulations
 - 32 merged PRs across p5.js core, web editor and website, including GPU-to-CPU transfer via storage buffers and WebGL visual regression test infrastructure
- CNCF – KubeStellar** Apr 2026 – May 2026
Distributed Systems & Automation
 - Reached 3rd highest contributor by systematically auditing the console and filing 500+ verified bug reports covering 118n gaps, silent failures and broken state handling
 - Resolved a race condition causing data loss across server restarts by migrating in-memory state to persistent Kubernetes ConfigMaps
 - Shipped 12 merged PRs delivering monitoring presets for Harbor Registry, Knative, Fluid, in-toto and Notary
- LFDT – Hiero SDK** Apr 2026 – Present
Automation & Tooling
 - Built hiero-automate, a config-driven PR automation pipeline with merge-conflict detection and ESLint integration
 - Shipped 15 merged PRs across hiero-sdk-cpp and hiero-website, covering build tooling, CI fixes and SDK maintenance

PROJECTS

- Split-it-Wise** | *Next.js · Firebase · Firestore* GitHub | Live
 - Built and deployed a group expense splitter now used daily by around 20 people across my hostel floor for real-money cost tracking
 - Handled all money maths in integer paise with unit tests, so balances always sum to zero with no float rounding drift, and enforced every permission server-side in Firestore rules rather than client code
 - Currently building the mobile version with React Native and a UPI deep-link settle-up flow
- DocRAG** | *Python · ChromaDB · Sentence-Transformers · Gemini* GitHub
 - Built a retrieval-augmented QA system over technical PDFs, fusing BM25 and vector search with min-max normalized weighted scoring for better recall on exact identifiers
 - Answers are generated strictly from retrieved context with page-level citations, and a comparison view exposes vector-only, BM25-only and hybrid retrieval side by side with a live fusion weight
- File-Lens** | *Node.js · Express* GitHub | Live
 - Built and deployed a storage analyzer where users pick a local folder straight from the browser and get instant insights on file size distribution, categories, and modification history, with only metadata reaching the server

PUBLICATIONS

- Processing Foundation:** “Drawing a Forest in One Line: A Preview of Instancing in p5.strands”, Aug 2026

ACHIEVEMENTS

- Processing Foundation OSS Micrograntee '26:** Paid grant to implement the WebGPU instancing API for p5.strands, nominated by project lead Kit Kuksenok from ~120 GSoC applicants
- CNCF KubeStellar – Admiral Badge:** Awarded for high-volume, high-impact contribution to the KubeStellar project

TECHNICAL SKILLS

- Languages:** JavaScript, Python, Go
- AI/ML & LLMs:** NumPy, Pandas, Matplotlib, Seaborn, LLM API Integration, RAG, Supervised Learning, Neural Network Fundamentals
- Frameworks/Libraries:** React, Next.js, Node.js, Express.js, p5.js, Streamlit
- Systems & Automation:** Kubernetes, Docker, Linux, Git
- Databases & API:** Firebase/Firestore, MongoDB, MySQL, ChromaDB, REST APIs
- Graphics/GPU:** WebGL, WebGPU, GLSL/WGSL